

Integrated Pandas Practical 2

Employee Performance & Payroll Analysis

Student Assignment - No Answers

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Scenario: You are a junior HR data analyst. HR has given you an employee master file containing missing values and suspected duplicate records. Management wants a clean dataset and a summary of salary and performance by department and city.

Chapters covered: Chapter 1 Sorting, Chapter 2 Grouping, Chapter 3 Missing Values, and Chapter 4 Duplicate Handling.

Dataset: employee_performance_payroll_practical.csv

Student instruction: This file contains questions only. Decide which Pandas method is appropriate, show your code and output, and add short interpretations for the management section.

1. Business Dataset

```
import pandas as pd
```

```
df = pd.read_csv("employee_performance_payroll_practical.csv")
```

Emp_ID	Name	Department	City	Experience	Salary	Rating
E101	Aarav	IT	Surat	3	45000	4.2
E102	Diya	HR	Mumbai	5	52000	4.5
E103	Rohan	Sales	Surat	2	38000	3.8
E104	Priya	IT	Ahmedabad		60000	4.7
E105	Arjun	Sales	Mumbai	6		4.1
E106	Neha	Finance	Surat	4	55000	4.3
E107	Rahul	HR	Ahmedabad	2	41000	3.9
E108	Kavya	IT	Mumbai	7	72000	4.8
E109	Vivek	Finance	Ahmedabad	5	61000	4.4
E110	Meera	Sales	Surat	3	42000	4.0

2. Part A - Initial Inspection

1. Load the CSV into df.
2. Display the first eight rows.
3. Find the number of rows and columns.
4. Display all column names.

5. Inspect data types and non-null counts.
6. Write two observations about possible data-quality issues.

3. Part B - Missing Values

1. Count missing values in each column.
2. Display rows containing at least one missing value.
3. Find the total number of missing cells.
4. Treat the missing Experience value using a sensible numerical summary and explain your choice.
5. Fill the missing Salary using the median Salary of the Sales department.
6. Verify that no missing values remain.

4. Part C - Duplicate Handling

1. Count later exact duplicate rows.
2. Display all rows involved in exact duplication.
3. Check repeated Emp_ID values using subset.
4. Explain why repeated Department or City values are not duplicate employee records.
5. Remove confirmed exact duplicates and reset the index.
6. Verify that no exact duplicates remain.
7. Report the cleaned number of employees.

5. Part D - Sorting

1. Sort employees by Salary from highest to lowest.
2. Display the five highest-paid employees.
3. Display the five employees with the highest Rating.
4. Sort first by Department and then by Salary from highest to lowest within each department.
5. Find the three least-experienced employees.
6. Find the highest-paid employee.

6. Part E - Grouping

1. Count employees in each Department.
2. Calculate average Salary by Department.

3. Calculate minimum, maximum, and average Salary by Department.
4. Calculate average Rating by Department.
5. Calculate average Salary by City.
6. Find maximum Salary in each Department.
7. Group by both City and Department and calculate average Salary.
8. Identify the Department with the highest average Salary.
9. Identify the Department with the highest average Rating.

7. Part F - Management Questions

1. Which department has the largest number of employees?
2. Which department has the highest average Salary?
3. Which department has the highest average Rating?
4. Which city has the highest average Salary?
5. Who is the highest-paid employee?
6. Who are the top three employees by Rating?
7. How many employees have a Rating of 4.5 or above?
8. Display employees earning more than 60,000, sorted from highest to lowest Salary.
9. Which department appears strongest on both salary and rating? Support your answer with numbers.
10. Why should HR inspect duplicate Emp_ID values before deleting records automatically?

8. Submission Checklist

- ☐ All questions attempted.
- ☐ Code and outputs included.
- ☐ Cleaning decisions explained.
- ☐ Duplicate records inspected before removal.
- ☐ Final cleaning verified.
- ☐ Management interpretations written in clear language.